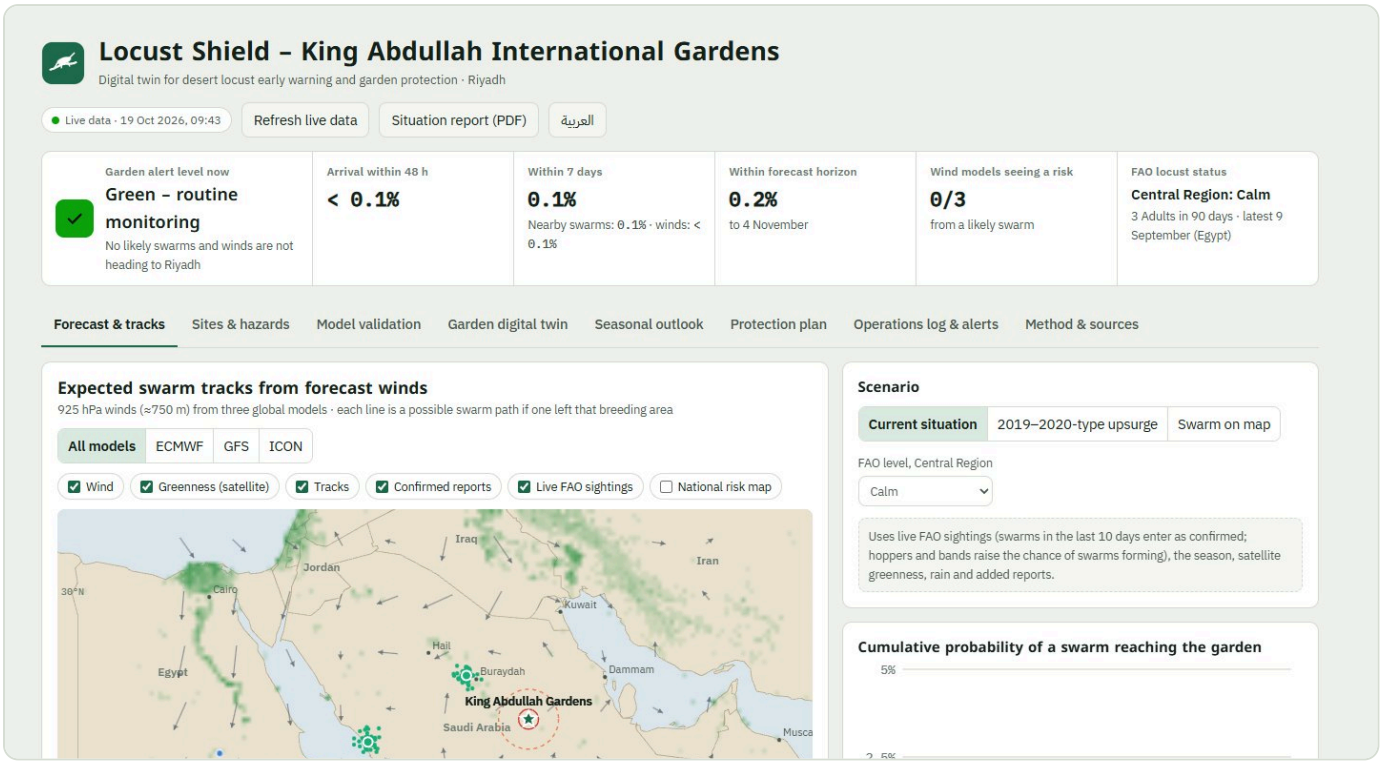
Locust Shield

User guide · Locust early-warning digital twin – King Abdullah International Gardens



About the program

Locust Shield forecasts the arrival of desert locust swarms from Africa and Yemen at King Abdullah International Gardens in Riyadh days in advance, and helps the garden prepare and protect itself before, during and after an invasion. It runs on real data refreshed daily: winds from three global weather models, rainfall at locust breeding areas, NASA satellite greenness, and bulletins of the UN Food and Agriculture Organization (FAO).

How to open it

- **On the computer:** double-click `LocustShield.exe` in the "درع الجراد" folder.
- **From any device or phone (no download):** open `https://asfantrading-create.github.io/locust-shield/?lang=en`
- **To download the program:** `https://github.com/asfantrading-create/locust-shield/releases/latest/download/LocustShield.zip`
- **Language:** the "العربية" button at the top switches the program to Arabic.
- **Data:** refreshes automatically. The green "Live data" badge means it is current.
- **Situation report:** the "Situation report (PDF)" button at the top produces a formal report to print or save as PDF for the authorities.

The top bar (always visible)

Sums up the situation at a glance: the garden's alert level, the chance of a swarm arriving within 48 hours, 7 days and the full forecast period (under the 7-day figure: closeness to existing swarms and wind tracks), how many wind models see a risk, and the latest live FAO sightings in the region.



Green
Routine monitoring

Yellow
Caution: low chance or season approaching

Orange
Threat: arrival possible within a week

Red
Danger: arrival likely within 48 hours

The tabs

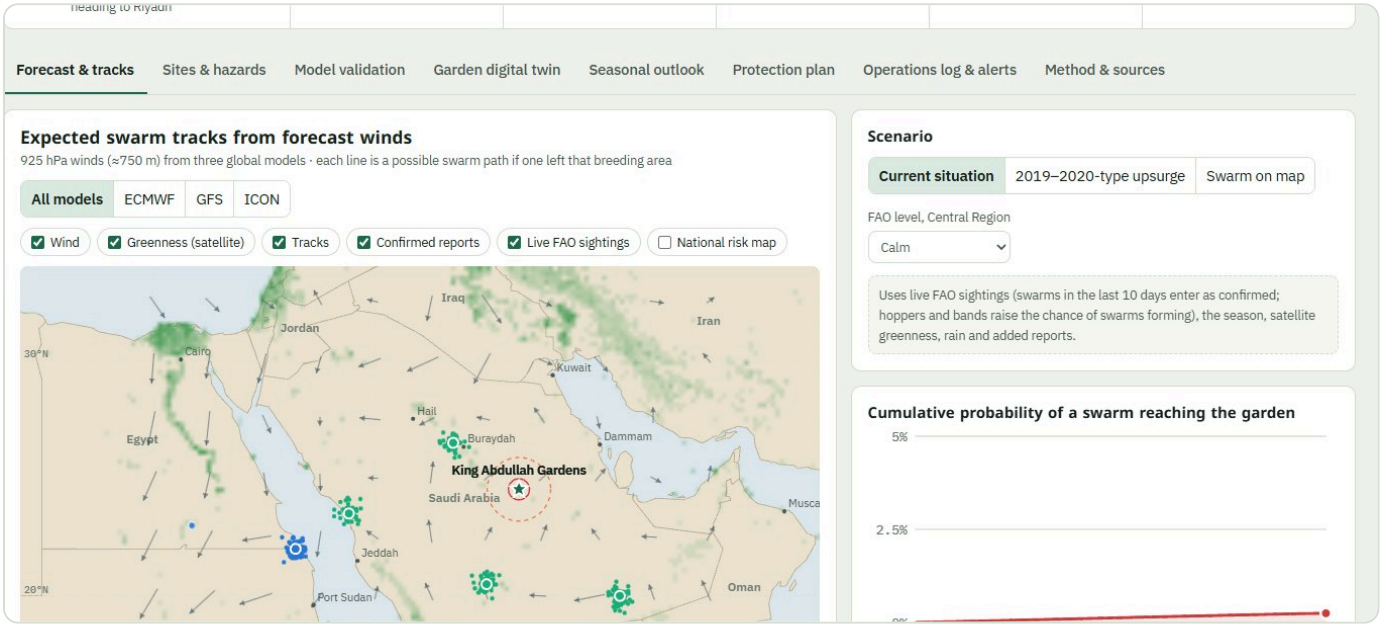
1 Forecast & tracks

What it shows

A map of current winds and possible swarm paths from 12 breeding areas, with live FAO locust sightings (swarms, hopper bands, hoppers, adults), a national risk map, and the chance of reaching the garden over the next 16 days.

How to use it

- Press "▶ Play" to watch swarms move day by day; switch on "National risk map" to see the most exposed areas.
- The FAO level is suggested automatically from live sightings and can be changed by hand.
- At the bottom: add a field report with a photo and notes, or import/export reports as Google Earth (KML), GeoJSON or CSV.



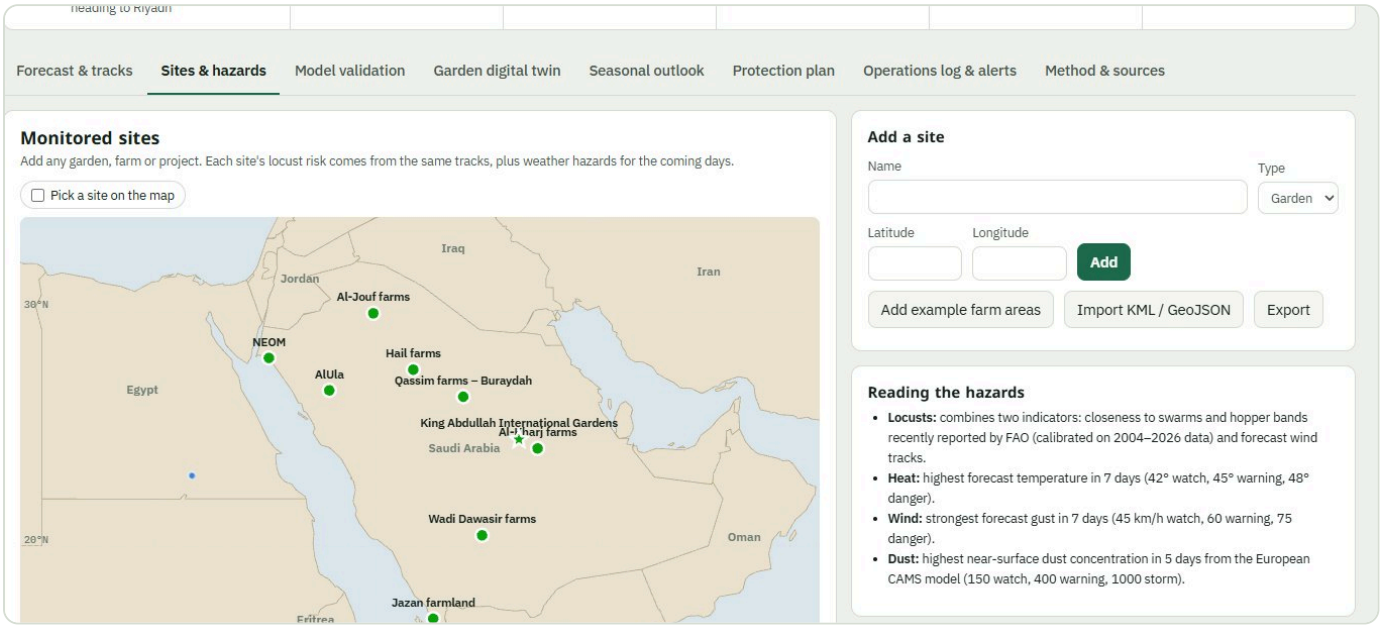
2 Sites & hazards

What it shows

All the sites you monitor (gardens, farms, projects) ranked by risk: locust risk per site, the nearest reported swarm, and weather hazards for the coming days – extreme heat, strong winds and dust storms.

How to use it

- Add a site by name and coordinates, or switch on “Pick a site on the map” and click it.
- “Add example farm areas” adds major Saudi farming areas, which you can delete.
- Import many sites at once from a Google Earth (KML) or GeoJSON file.



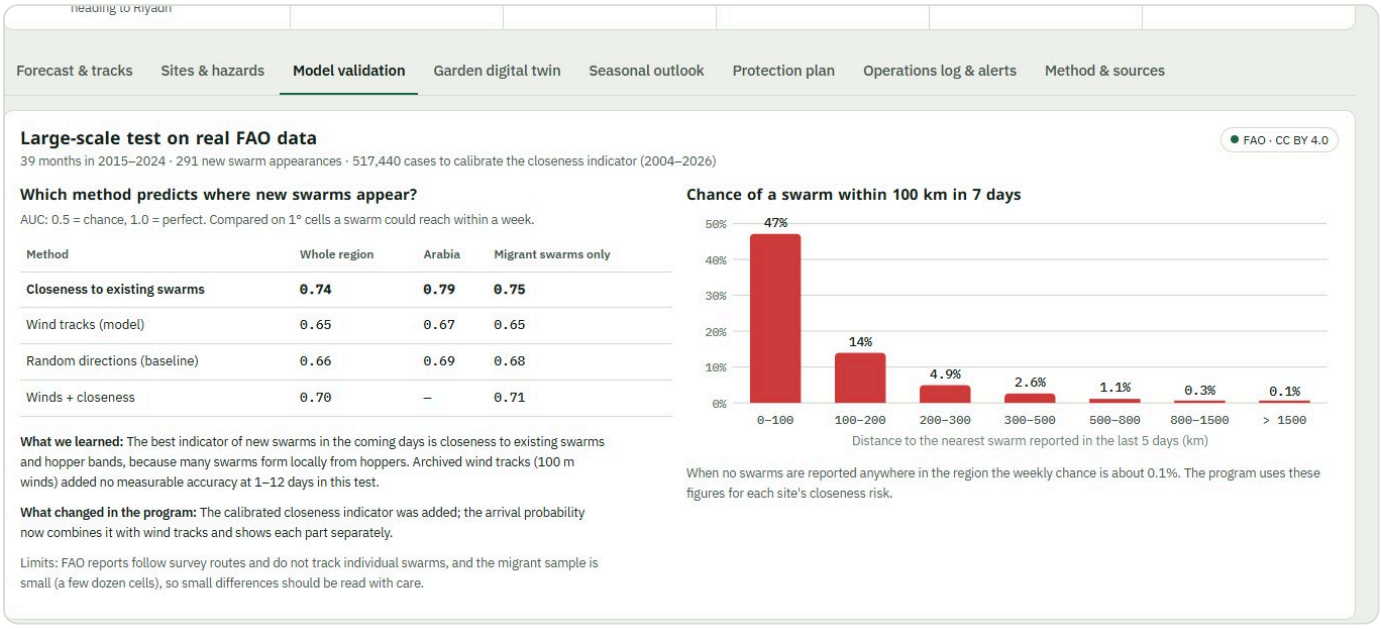
3 Model validation

What it shows

Scientific evidence of accuracy: a large-scale test on hundreds of real swarms in FAO data (2015–2024) comparing forecasting methods, the closeness curve calibrated on 2004–2026 data, and replays of four documented invasions.

How to use it

- Read the comparison table and the “chance of a swarm within 100 km in 7 days” curve at the top.
- Pick a historical event and press “▶ Replay” to see the tracks and where locusts were actually reported.



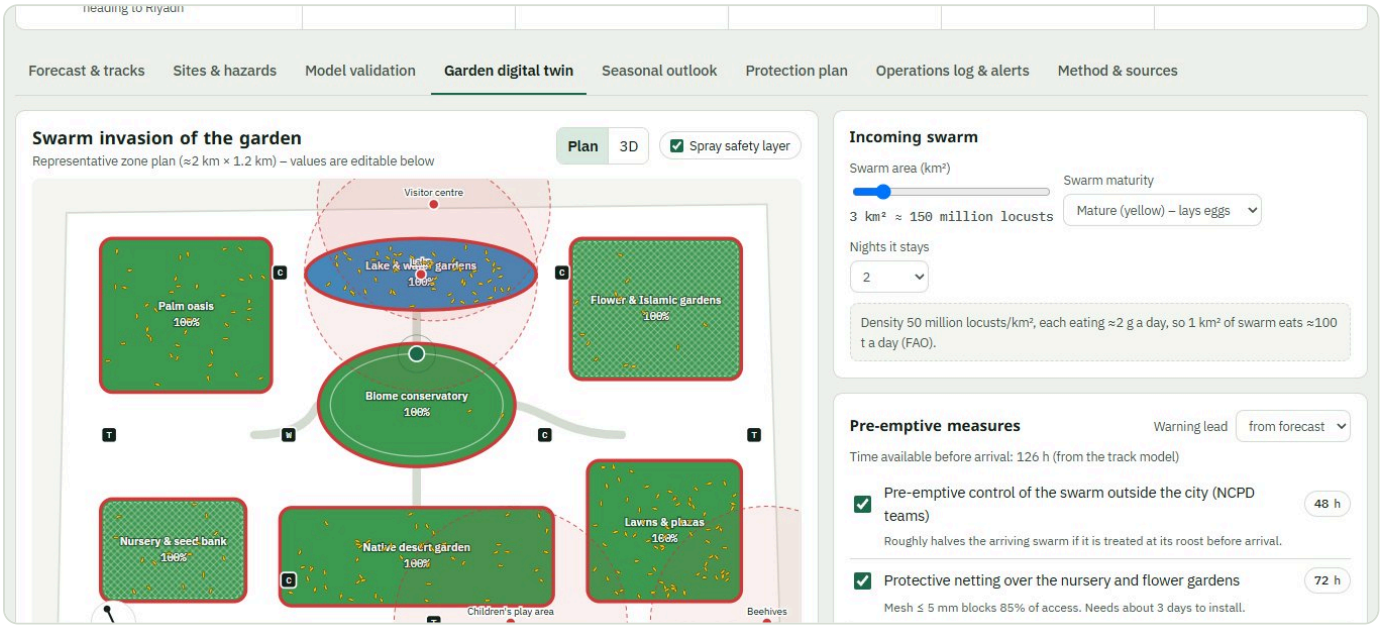
4 Garden digital twin

What it shows

An hour-by-hour simulation of a swarm landing on the garden: what it eats in each zone, the estimated loss in riyals, recovery time, and the effect and lead time of each protective measure.

How to use it

- Set the swarm size and maturity, then press "▶ Run invasion".
- Switch measures on or off and compare the results and the cost-benefit table.
- The "Spray safety layer" shows where spraying is allowed given the dawn wind; further down are the post-invasion timeline and the egg-hatching date.



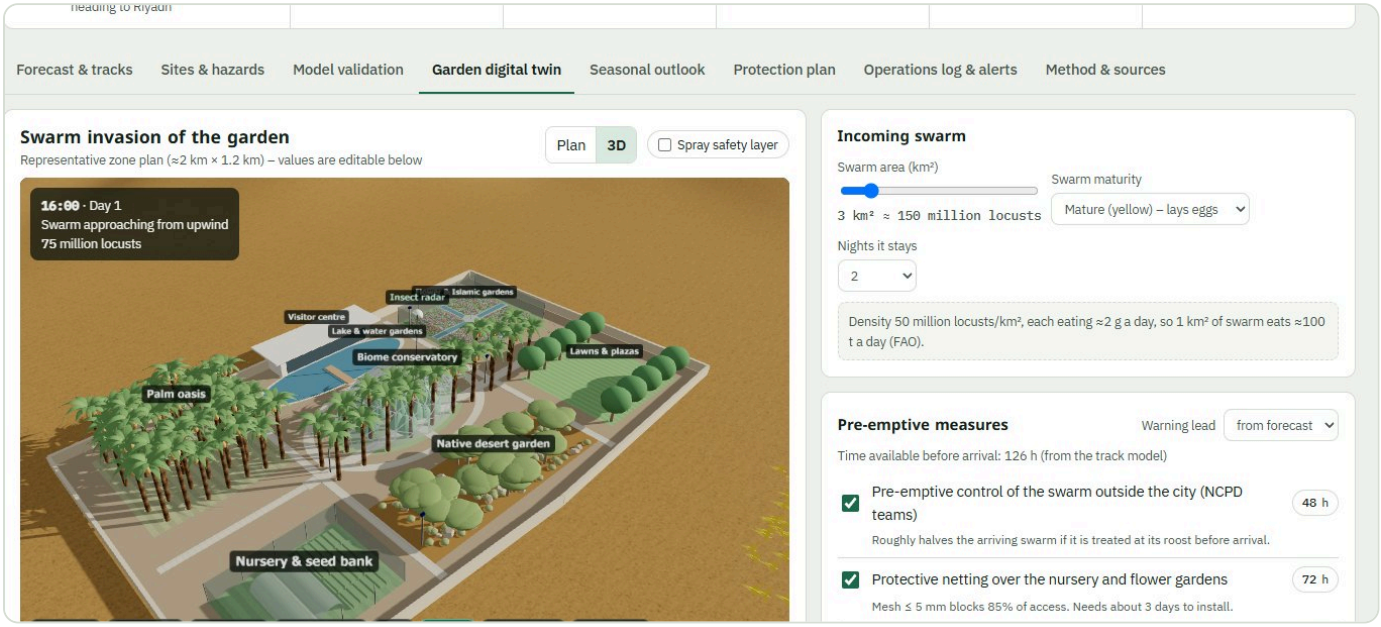
3D 3D view (inside the digital twin)

What it shows

A 3D model of the garden – palm oasis, glass conservatory, nursery, desert garden – with the swarm approaching, feeding, and roosting in the trees at night. Lighting follows the simulation clock.

How to use it

- Press “3D” in the digital twin tab.
- Drag to rotate, scroll to zoom, or use the preset camera buttons.
- “Full screen” works well for presenting in meetings.



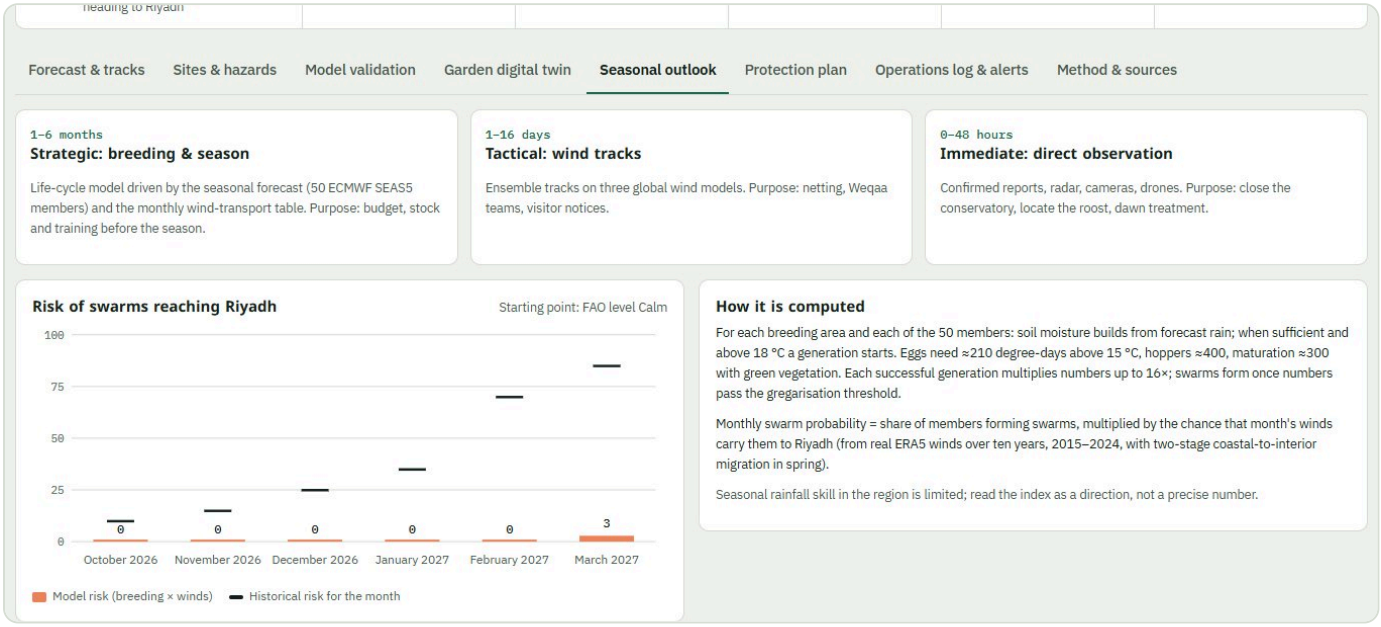
5 Seasonal outlook

What it shows

The risk of swarms reaching Riyadh over the next six months, from the seasonal rain and temperature forecast, the locust life cycle and ten years of winds.

How to use it

- Bars show the forecast risk for each month; the dark mark shows the historical risk for that month.
- The table shows which breeding areas may form swarms. Risk is usually highest from February to May.



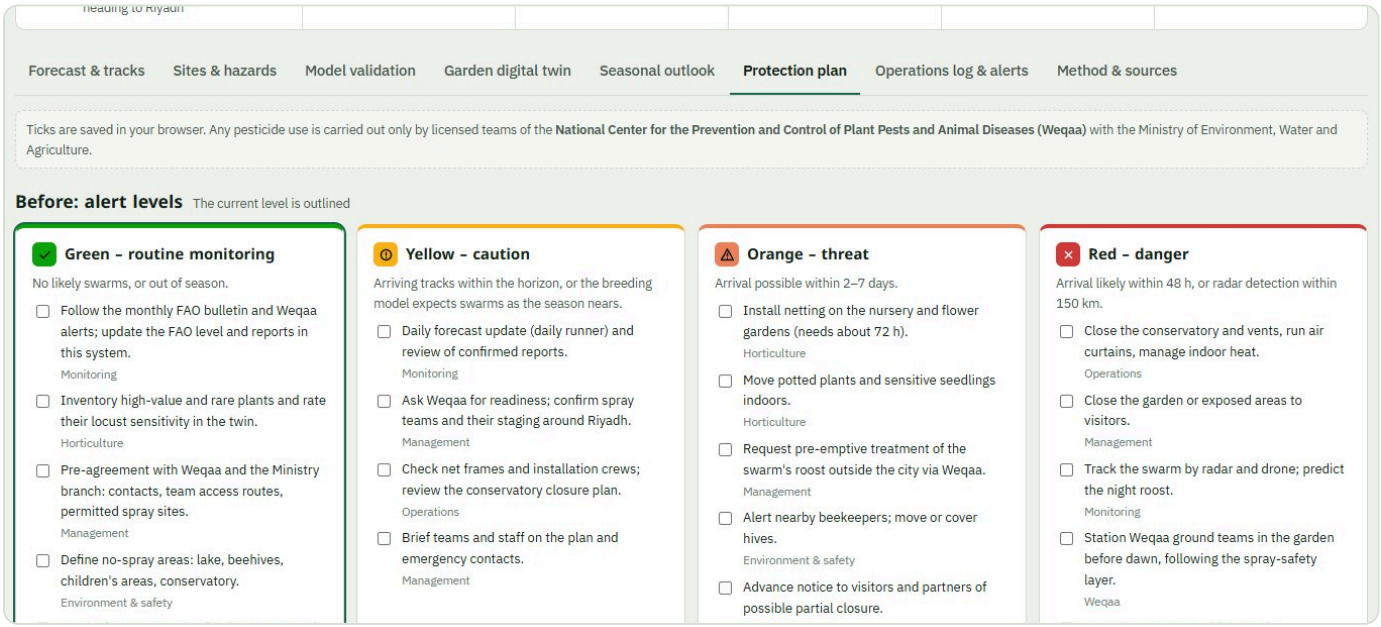
6 Protection plan

What it shows

Ready task lists for each alert level (green, yellow, orange, red), then what to do during and after an invasion, with the team responsible for each task.

How to use it

- The current level is outlined.
- Tick completed tasks to track readiness.



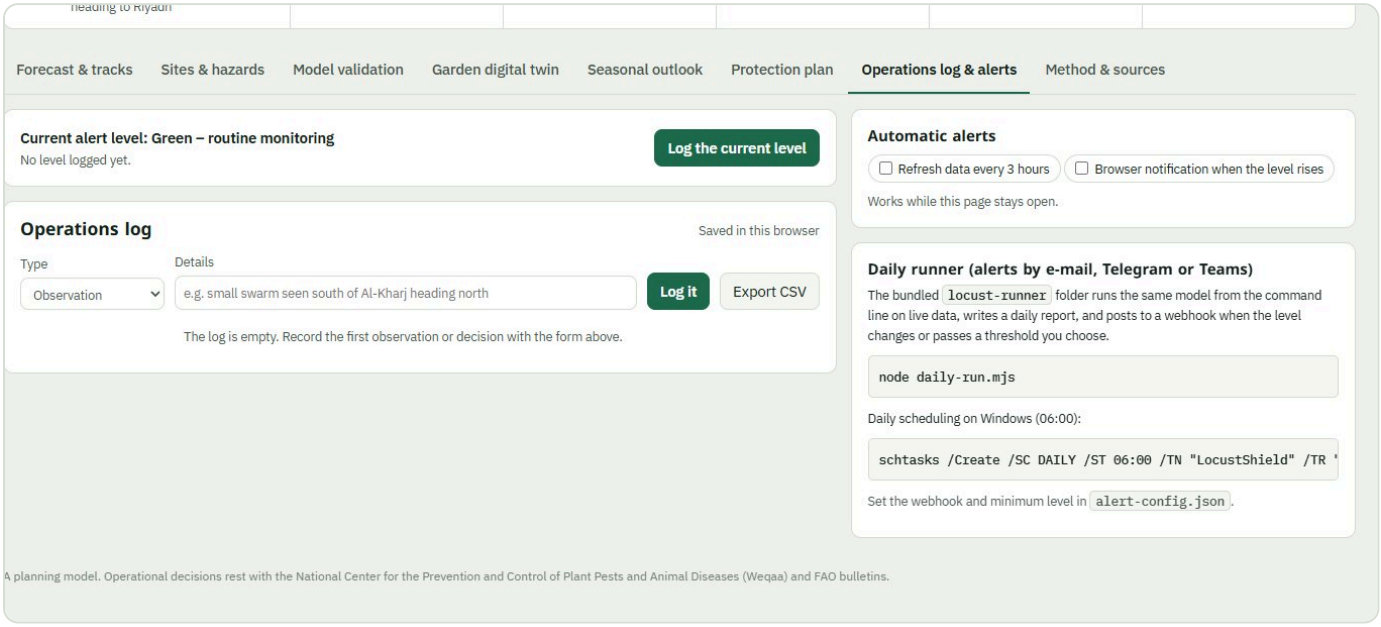
7 Operations log & alerts

What it shows

A shared team log for reports, decisions, actions taken and alert-level changes, with export and alerts.

How to use it

- Choose a type, write the details and press “Log it”.
- “Export CSV” saves the log for Excel.
- The daily runner sends an alert to Teams or Telegram when the risk rises.



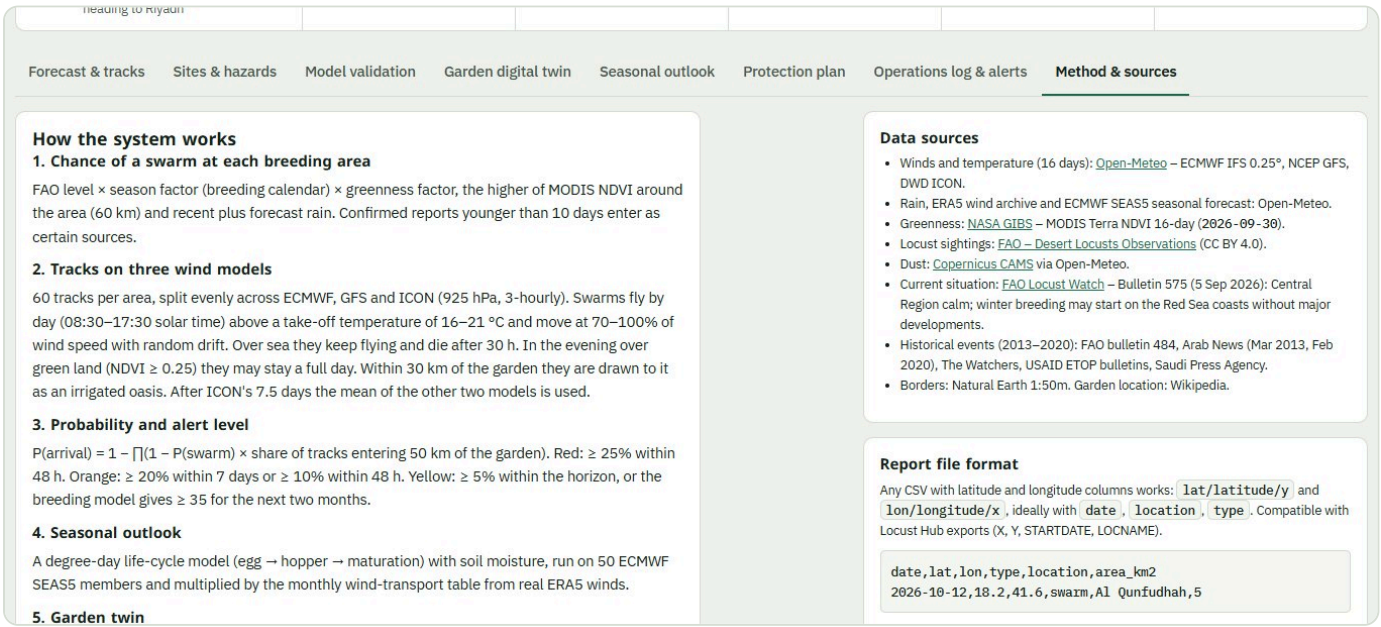
8 Method & sources

What it shows

A plain explanation of how the model works and its limits, and the data sources: FAO, global weather models and NASA satellites.

How to use it

- Your reference for “where do these numbers come from?”



Important notes

- The program supports planning and early warning; operational decisions and any pesticide use rest with the National Center for the Prevention and Control of Plant Pests (Weqaa) and FAO bulletins.
- If live data cannot be fetched (for example without internet), the program uses its last saved data and shows a yellow bar with its date.
- The garden plan and its values are representative and can be edited in the “Garden zones and values” table.